Memory Physics, Finite Capacity, and Topology in Heterogeneous Quantum Networks

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Abstract

Quantum-network capacity planning couples memory physics, finite storage, topology, and traffic. We study these interactions using a discrete-event simulator with six representative memory profiles. A memory–topology atlas contains 9,360 completed runs, complemented by 8,730 confirmatory runs and an extended per-node memory-budget sweep. We distinguish a threshold relative to the best performance observed on a finite grid from a physical saturation limit. Among 840 reported threshold estimates, 22.3% reach 95% of the observed maximum by four slots per node, while 35.5% first reach it at the upper grid boundary of 64 slots. Four tested scalar reductions do not provide a topology-independent collapse; the best pooled quadratic fit explains 37.9% of response variance. The retrieval-window parameter $\chi_r = \tau_r p_{\mathrm{gen}}/t_{\mathrm{attempt}}$ provides an interpretable descriptor, but an independent-channel approximation does not establish a network memory-sizing law. In 5,310 heterogeneous-placement runs, hub and betweenness-based placement outperform peripheral placement in aggregate, with topology-specific exceptions. Coupling assumptions change absolute performance while preserving the aggregate strategy ordering. Harmonized model comparisons provide limited robustness evidence, not independent native-protocol validation. The results support model-specific capacity and placement analysis with explicit metric definitions, finite-grid limitations, and interface assumptions.

Keywords: quantum networks; quantum memory; discrete-event simulation; finite capacity; heterogeneous placement; reproducibility

1 Introduction

Quantum networks promise distributed quantum computation, secure communication, and distributed sensing [1]. A central engineering question is not simply *which memory technology is best*, but rather: *how do physical memory properties, finite capacity, network position, topology, and offered load jointly determine useful end-to-end entanglement?* This question couples device-level physics (coherence time, retrieval efficiency, readout fidelity, physical interfaces) to network-level structure (topology, routing, contention), and its answer determines hardware–network co-design rules for heterogeneous quantum networks.

Existing analytical results characterize scaling and rate–memory trade-offs for specific repeater architectures, often homogeneous chains with prescribed memory models [2, 3, 4]. Simulation frameworks such as SeQUeNCe [5], NetSquid [6], and QuISP [7] enable more detailed studies, while recent cross-validation work shows that simulator assumptions can materially affect timing even when fidelity agrees [8]. Recent work on heterogeneous link architectures [9] and hybrid repeater designs [10, 11] motivates the question of whether conclusions derived from homogeneous models transfer to heterogeneous hardware.

Simulation-based network-design conclusions are valuable only if they are not artifacts of a single implementation. We therefore supplement the component-level analytical validation with a harmonized cross-implementation robustness check (Sec. 4.3), in which selected conclusions are re-evaluated through harmonized implementations using SeQUeNCe and SimQN adapters. The purpose is to test whether selected structural conclusions survive a harmonized model pathway with shared code, not to establish universal simulator invariance. A companion reliability study reports the full simulator-invariance methodology.

This paper addresses four research questions:

RQ-A1 Which memory- and network-level parameters limit end-to-end entanglement delivery?

RQ-A2 How does performance vary across representative memory profiles and topologies?

RQ-A3 Does a universal scaling law for memory sufficiency exist?

RQ-A4 How does heterogeneous memory hardware affect conclusions?

Our central claim is that memory sufficiency and heterogeneous placement are conditional on the modeled physics, topology, traffic, and interface assumptions. We test specific scalar descriptions; their failure is not a proof that no useful scaling relation can exist.

We distinguish the retrieval-window descriptor

$$\chi_r = \frac{\tau_r p_{\text{gen}}}{t_{\text{attempt}}} \tag{1}$$

from $\chi_2 = T_2 p_{\text{gen}}/t_{\text{attempt}}$, which describes a coherence-time scale. Retrieval loss and decoherence affect different parts of the delivery process. For the six adopted bundles the two lifetimes have the same rank ordering, but they are not proportional. This correlation does not identify a causal effect or validate a universal memory-sizing law. Section 8 gives a restricted analytical interpretation.

2 Related Work

2.1 Quantum Repeater Scaling

Scaling behavior for quantum repeaters has been analyzed under a range of homogeneous-chain assumptions. Razavi, Piani, and Lütkenhaus [2] showed that finite memory coherence can change favorable polynomial distance scaling into an exponential degradation in $\sqrt{L}$ in their optimized repeater model. More recent work studies exact rate analysis with imperfect memories [12] and joint optimization of repeater count, spacing, and distillation under rate and fidelity constraints [4]. These results provide important benchmarks, but they do not directly address the cross-product of heterogeneous memory profiles and network topology studied here.

2.2 Topology and Entanglement Distribution

The impact of network topology on entanglement distribution has been studied by several groups. Ooi, Sutcliffe, and Beghelli [13] identified four topology-dependent performance regimes across 81 real topologies. Ozaydin et al. [14] developed optimal algorithms for multi-ring topologies. Vardoyan et al. [15] derived end-to-end capacity bounds via flow maximization. Pirandola [16] derived end-to-end capacity bounds for quantum networks. Takeoka et al. [17] established the fundamental rate-loss tradeoff for optical quantum communication, constraining the achievable rates that any network design must respect. Vasantam and Towsley [18] analyzed the stability of quantum networks with max-weight scheduling, providing a queueing-theoretic foundation for resource contention analysis. These studies focus on routing, capacity, and scheduling rather than memory-physics interactions.

2.3 Heterogeneous Quantum Networks

Recent work has begun exploring heterogeneous hardware. Soon et al. [9] demonstrated that mixed link architectures work without significant rate loss using QuISP. Sun et al. [10] and Cussenot et al. [11] proposed hybrid repeater architectures combining trapped ions, neutral atoms, and AFC memories. These studies motivate, but do not replace, a controlled comparison of representative memory profiles across multiple topology classes and placement rules.

2.4 Resource Placement

Repeater and memory placement has been formulated as an optimization problem. Pouryousef et al. [19] developed a network-planning framework that allocates repeaters and memories while accounting for rate, fidelity, multiplexing, and coherence. Sripotchanart et al. [20] proposed a scalable two-step linear-programming approach for repeater placement, while Islam and Arslan [21] developed fast heuristic deployment models. Our focus differs: we hold the network-planning objective fixed and study how heterogeneous memory quality, finite capacity, topology, and interface assumptions interact.

2.5 Simulator Validation

Cross-validation of quantum network simulators is an emerging concern. Chung et al. [8] cross-validated QuISP and SeQUeNCe, finding that different simulator assumptions can produce different timing results even when fidelity results agree after harmonization. This motivates careful validation of any single-simulator study. The present work supplements its component-level analytical validation with a harmonized cross-implementation robustness study (Sec. 4.3). We emphasize that this cross-implementation check does not yet use native SeQUeNCe or SimQN protocols, does not include NetSquid or QuISP, and that the SeQUeNCe and SimQN adapters share a common harmonized model. The purpose is therefore to test whether selected conclusions survive a harmonized model pathway with shared code, not to establish universal simulator invariance.

3 Quantum Memory and Network Model

3.1 Representative Memory Profiles

We model six representative quantum-memory profiles using literature-informed parameter bundles (Table 1):

Table 1: Representative memory-profile parameters. T_1 and T_2 are in seconds, η_r is retrieval efficiency at $t = 0$, τ_r is the retrieval decay time (s), and F_{ro} is readout fidelity. Adopted-value provenance and contextual source citations are provided in Supplemental Table S1 [22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33].

Profile	T_1 (s)	T_2 (s)	η_r	τ_r (s)	F_{ro}
Trapped Ion	100	10.0	0.90	10.0	0.999
Neutral Atom	1.0	0.1	0.80	0.5	0.99
Diamond (Electron)	1.0	0.044	0.70	0.1	0.95
Diamond (Nuclear)	100	1.3	0.50	1.0	0.95
AFC	10	0.005	0.50	0.01	0.99
Superconducting	0.001	0.0001	0.95	0.001	0.999

These profiles are *representative modeling bundles*, not complete device-level hardware comparisons. Some bundles combine characteristic values reported for related implementations within a memory platform; Supplemental Table S1 records the adopted values and separates literature

context from direct numerical provenance. Values without a verified source location are treated as modeling assumptions. The simulated budget m is an abstract number of independently usable memory slots per node and is not identified with an experimentally reported multimode capacity.

3.2 Network Topologies

We study nine topology classes: chain, star, random geometric graph (RGG), Waxman, hierarchical, lattice, scale-free, binary tree, and ring. Topologies are generated with controlled parameters (node count, edge density, geographic extent).

3.3 Simulator

Our discrete-event simulator models:

- Per-node memory capacity via a *placement* dictionary
- Homogeneous or heterogeneous memory profiles
- Homogeneous or heterogeneous decoherence models
- Depolarizing, dephasing, or amplitude-damping noise
- Serial or parallel link generation
- Entanglement swapping with configurable success probability
- Retrieval efficiency and readout fidelity
- Deadline-based service qualification
- Poisson request arrivals with optional traffic matrices
- Warmup and censoring to reduce transient artifacts

Table 2 lists the default simulator parameters.

Table 2: Simulator parameters used throughout the paper unless otherwise stated.

Parameter	Value	Description
p_{gen}	0.005	Link generation success probability
t_{attempt}	10^{-3} s	Link attempt interval
Deadline	1.0 s	Request deadline
$F_{\min}$	0.5	Minimum fidelity for qualification
p_{swap}	0.98	Entanglement swap success probability
F_{initial}	0.9	Initial link fidelity
T_{obs}	200 s	Simulation observation time
Warmup	First 10% of requests	Warmup period
n_{requests}	100	Requests per simulation

Noise levels are defined for the sensitivity analysis (Sec. 10) as: **low** = 0.1× baseline depolarizing rate, **medium** = 1.0× baseline, **high** = 10× baseline. The baseline depolarizing rate is derived from the memory profile's T_2. In the atlas and confirmatory experiments, noise is not independently varied; the depolarizing rate is fixed by each profile's T_2.

Run-length caveat: With $n_{\text{requests}} = 100$ and the highest traffic level $\lambda = 50$, the observation window covers approximately $100/50 = 2$ s of simulated time. This is short relative to the $T_{\text{obs}} = 200$ s parameter and may not reach full stationarity for queueing dynamics at high load. The warmup period (first 10% of requests) partially mitigates transient effects, but the high-traffic results should be interpreted with the caveat that the system may not have fully reached steady state.

The primary performance metric is A_{req}: the fraction of *effective incoming requests* (excluding warmup, censored, and no-path requests) that are completed within deadline. A companion metric A_{qual} additionally requires the final fidelity to exceed $F_{\min}$. Quality-qualified throughput showed stronger ordinal agreement across the tested harmonized implementations than completion-only throughput. Because request-processing and denominator conventions were not fully identical across implementations, we treat this as a robustness observation rather than as a general superiority claim for either metric. All confidence intervals reported in this paper are 95% bootstrap intervals (10,000 resamples, percentile method) computed across seeds unless otherwise stated.

Definition of m: Throughout this paper, m denotes the *per-node memory budget*: every node is assigned m memory slots. This definition is used consistently in the atlas, the confirmatory replication, and the m_c audit.

3.4 Mathematical Model

The simulator's quantum-state and noise models are defined as follows.

Werner-state parameterization. Each entangled link is represented as a Werner state with fidelity F to the target Bell state $|\Phi^+\rangle$:

$$\rho = F\,|\Phi^+\rangle\langle\Phi^+| + \frac{1-F}{3}\sum_i |\Psi_i\rangle\langle\Psi_i|, \tag{2}$$

where $\{|\Psi_i\rangle\} = \{|\Phi^-\rangle, |\Psi^+\rangle, |\Psi^-\rangle\}$ are the other three Bell states. The Werner state is entangled iff $F > 0.5$.

Decoherence channels. During memory storage, the depolarizing channel drives F toward the completely mixed value $F = 0.25$:

$$F(t) = 0.25 + (F_0 - 0.25)\,\exp(-\Gamma\, t), \tag{3}$$

where Γ is the depolarizing rate derived from the memory profile's T_2. The simulator also supports pure dephasing and amplitude-damping channels; in the atlas and confirmatory experiments, the depolarizing channel is used with Γ determined by T_2.

Entanglement-swap composition. Multi-hop entanglement is established via Bell-state-measurement (BSM) swaps with success probability p_{swap}. The simulator composes Werner states via the Werner parameter $w = (4F-1)/3$. For two links with Werner parameters w_1 and w_2 and a BSM with Werner parameter $w_{\text{bsm}} = (4F_{\text{bsm}} - 1)/3$, the output after a successful swap is

$$w_{\text{new}} = w_1\, w_2\, w_{\text{bsm}}, \qquad F_{\text{swap}} = \frac{3\, w_{\text{new}} + 1}{4}, \tag{4}$$

where F_{bsm} is the BSM fidelity (default $F_{\text{bsm}} = 0.99$). For an ideal BSM ($F_{\text{bsm}} = 1$, i.e., $w_{\text{bsm}} = 1$), this reduces to the standard Werner-state swap formula

$$F_{\text{swap}} = F_1 F_2 + \frac{(1-F_1)(1-F_2)}{3}. \tag{5}$$

The swap succeeds with probability p_{swap} per hop. Failed swaps consume the stored pairs.

Retrieval efficiency. When a stored pair is retrieved from memory, the retrieval succeeds with probability given by an exponential decay model

$$\eta(t) = \eta_r \exp\left(-\frac{t}{\tau_r}\right), \tag{6}$$

where η_r is the profile-dependent retrieval efficiency at $t = 0$ and τ_r is the retrieval decay time (both listed in Table 1). The simulator also supports a Gaussian model $\eta(t) = \eta_0 \exp[-(t/\tau)^2/2]$ for AFC-type memories, but the exponential model is used throughout the atlas and confirmatory experiments. The frozen simulations apply a multiplicative readout-fidelity rule:

$$F_{\text{out}} = F_{ro}F. \tag{7}$$

This phenomenological rule is the one relevant to the reported datasets; it is not identified here with a depolarizing quantum channel.

T1/T2 compatibility. All memory profiles satisfy the physical constraint $T_2 \leq 2\,T_1$, which follows from the relation $1/T_2 = 1/(2T_1) + 1/T_\phi$ with $T_\phi \geq 0$ (the pure dephasing time T_ϕ is non-negative). This is a constraint on the parameter space, not a derivation of a specific value; the adopted T_2 values in Table 1 satisfy this inequality, but the inequality alone cannot determine them.

Readout-model note. The frozen atlas, confirmatory, and m_c datasets are attributed to commit 1f86a98 (4 October 2026). A later code version uses $F_{\text{out}}^{\text{new}} = F_{ro}F + (1 - F_{ro})/4$. At a fixed pre-readout fidelity, the exact change is $(1 - F_{ro})/4$, or 0.00025–0.0125 for the listed profiles. This can change fidelity-qualified outcomes near a threshold. Completion-only outcomes remain unchanged only if the implementation does not use fidelity for acceptance, routing, or scheduling. Merely omitting F_{ro} from a scalar predictor does not establish invariance of its target. Reproduction must use the frozen rule and record the code revision.

4 Simulator Validation

4.1 Component-Level Analytical Validation

We validate the simulator's component models against analytical formulas for elementary link generation probability, Werner-state decoherence, multi-hop swapping fidelity composition, and retrieval efficiency/readout. The implementation is verified against the formulas (numerical error $< 10^{-10}$).

4.2 Component-Level Reference Comparison

As an additional reference-model check, we evaluate five scenario families (V1: elementary link, V2: three-node repeater, V3: multi-hop, V4: finite memory, V5: decoherence crossover) using parameterizations aligned with SeQUeNCe component models. This exercise is *not* a protocol-level cross-simulator validation because the SeQUeNCe protocol stack is not executed end-to-end. Accordingly, we use these results only as component-level validation and do not infer protocol-level agreement. Table 3 shows the component-level reference comparison.

Table 3 shows that fidelity agreement is moderate (median $|\Delta F| = 0.065$) but ranking correlation varies substantially by scenario, from $\rho = 0.32$ (V2) to near zero (V4, V5). The low ranking correlation in V4 and V5 reflects differences in how the two models handle finite-memory and decoherence effects. A matched protocol-level comparison against an independent codebase remains future work.

Table 3: Component-level reference comparison per scenario. ρ is the Spearman rank correlation between WestQuant and reference-model A_{req}.

Scenario	n	Median $\lvert\Delta F\rvert$	Median $\lvert\Delta A\rvert$	ρ
V1: Elementary link	270	0.017	0.178	0.218
V2: Three-node repeater	270	0.058	0.356	0.319
V3: Multi-hop	360	0.103	0.378	0.130
V4: Finite memory	270	0.152	0.356	0.027
V5: Decoherence crossover	540	0.042	0.244	0.025
Overall	1,710	—	—	—

4.3 Cross-Implementation Robustness Check

A cross-implementation robustness check was performed using harmonized implementations on the SeQUeNCe and SimQN event frameworks. In summary: the topology ordering (RGG > star > chain) and the memory-profile ordering (trapped ion > neutral atom > diamond electron > AFC) were reproduced under A_{qual}, while absolute performance values and finite-memory contention were implementation-sensitive. The SeQUeNCe and SimQN adapters share a common harmonized model and do not constitute two independent physical-model replications; native SeQUeNCe/SimQN protocols were not included.

5 Representative Memory-Profile Atlas

5.1 Experimental Design

The atlas spans 6 representative memory profiles, 9 topology classes, 3 traffic levels ($\lambda \in \{1, 10, 50\}$), 3 network sizes ($N \in \{10, 20, 50\}$), 4 memory budgets ($m \in \{1, 2, 4, 8\}$, per node), and 5 random seeds. Noise is not independently varied in the atlas; the depolarizing rate is determined by each memory profile's T_2 (Sec. 3). The full Cartesian design contains $6 \times 9 \times 3 \times 3 \times 4 \times 5 = 9{,}720$ nominal cells, of which 9,360 completed runs are retained. The 360 missing cells (3.7%) arise from disconnected random-geometric graphs at small N and from simulation failures; no missing cells are imputed. The frozen atlas dataset is `v05_fixes/results/p3_scalar_test.csv`.

5.2 Confirmatory Replication

A confirmatory campaign spanning 6 profiles, 5 topologies (chain, star, RGG, lattice, scale-free), 3 traffic levels, 5 budgets $m \in \{1, 2, 4, 8, 16\}$, fixed $N = 20$, and 20 seeds produced a frozen dataset of 8,730 completed simulations (frozen file: `v05_fixes/results/p2_c3_extended.csv`). The full Cartesian design contains $6 \times 5 \times 3 \times 5 \times 20 = 9{,}000$ cells; 270 cells (3.0%) are missing due to disconnected random-geometric graphs and simulation failures. No missing cells are imputed. The completed dataset confirms:

- **C1 (family effect):** Memory family materially affects A_{req} (Trapped Ion mean= 0.453 vs Superconducting mean= 0.003). The tested cross-implementation analysis reproduced the ordering of the evaluated memory profiles under the fidelity-qualified metric, whereas absolute performance differences were implementation-sensitive.
- **C2 (topology effect):** Topology materially affects A_{req} (RGG mean= 0.233 vs chain mean= 0.082). At the fixed $N = 20$ used here, equal per-node budgets also give equal total memory Nm. Path-length distributions and graph-selection rules still differ between

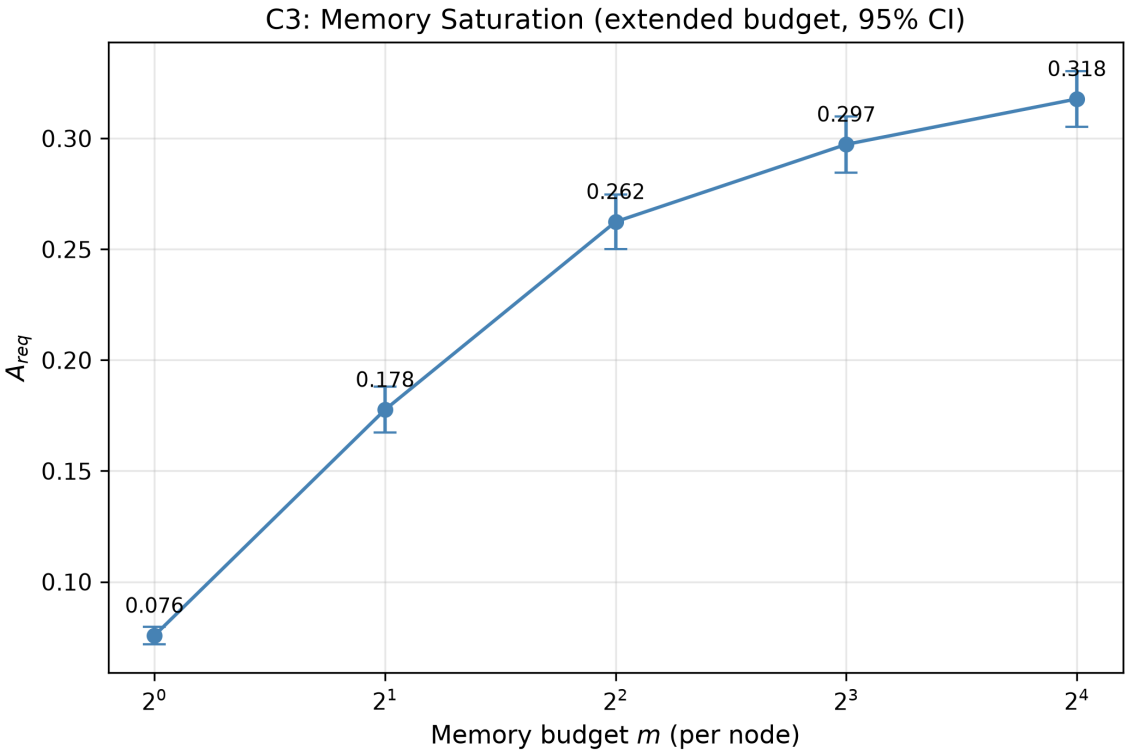


Figure 1: C3 memory-budget response with extended budget ($m = 1, 2, 4, 8, 16$, per node). Error bars show 95% CI across 20 seeds. The curve flattens but does not reach a clear plateau within the tested range.

topologies. The RGG > star > chain ordering was reproduced in the harmonized cross-implementation experiment.

- **C3 (diminishing returns):** Memory-budget response shows diminishing marginal gains but the 95%-of-observed-maximum threshold is not fully resolved at $m = 16$ ($m = 1$: 0.076, $m = 2$: 0.178, $m = 4$: 0.262, $m = 8$: 0.297, $m = 16$: 0.318). Monotonicity of $A(m)$ was reproduced in the cross-implementation check; the diminishing-returns conclusion rests on the finer WestQuant sweep and was not resolved by the coarser cross-implementation grid.

Figure 1 shows the extended budget-response curve. The curve is nondecreasing and the marginal gain diminishes, but a clear asymptotic plateau is not reached within $m \leq 16$.

Data provenance: All quantitative results in this paper are drawn from the frozen `v05_fixes` dataset directory, not from the earlier exploratory experiments in the `a07` or `a10` directories. The atlas uses `p3_scalar_test.csv` (9,360 runs), the confirmatory replication uses `p2_c3_extended.csv` (8,730 runs), and the m_c audit uses `p1_mc_values.csv` (840 m_c estimates per α). Earlier exploratory runs (`a07`: 10,080 raw results, 2,817 m_c estimates; `a10`: 5,238 confirmatory runs) used different parameter grids and are not the basis for the numbers reported here.

6 Memory-Budget Response and Memory-Budget Threshold

6.1 Budget-Response Curves

Figure 2 shows memory-budget response curves for four memory families across three topologies, using the per-node budget definition and an extended sweep $m \in \{1, 2, 4, 8, 16, 32, 64\}$.

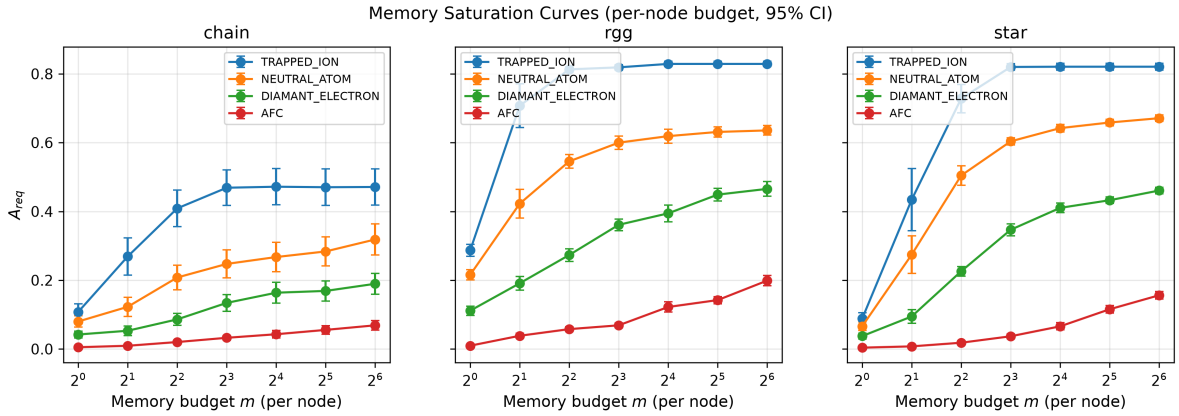


Figure 2: Memory saturation curves with 95% CI for four memory families across three topologies (per-node budget $m \in \{1, 2, 4, 8, 16, 32, 64\}$). Trapped-ion profiles approach a plateau within the tested range; AFC profiles are still increasing at $m = 64$.

6.2 Memory-Budget Threshold (m_c)

We define the memory-budget threshold as the smallest per-node budget at which the primary performance metric A_{req} reaches a specified fraction of the best observed performance on the tested budget grid:

$$m_c(\alpha) = \min \left\{ m : A_{\text{req}}(m) \geq \alpha \cdot A_{\max}^{\text{obs}} \right\} \tag{8}$$

with primary $\alpha = 0.95$. Here $A_{\max}^{\text{obs}} = \max_m A_{\text{req}}(m)$ is the maximum over the *observed* budget grid $m \in \{1, 2, 4, 8, 16, 32, 64\}$, not an asymptotic limit. Because $A_{\max}^{\text{obs}} > 0$ whenever any budget produces a nonzero signal, some budget on the grid always satisfies the criterion; the threshold is therefore always finite on the grid. We exclude regimes where $A_{\max}^{\text{obs}} < 0.02$ (no meaningful signal), The reported 5,880 completed budget runs correspond to 840 complete seven-budget curves; the 60 missing curves are accounted for by graph exclusions, not an additional set of 60 low-signal removals. A row-level exclusion ledger is needed to verify the low-signal filter separately. (The companion metric A_{qual} was used in the cross-implementation check because it showed stronger ordinal agreement across implementations, but the primary m_c definition uses A_{req}, consistent with the primary metric declared in Sec. 3.)

Three distinct concepts should not be conflated:

1. **Grid threshold:** The first budget on the discrete grid $\{1, 2, 4, 8, 16, 32, 64\}$ that reaches $\alpha \cdot A_{\max}^{\text{obs}}$. This is what m_c measures. A threshold at $m_c = 16$ does not imply that the minimum integer budget is exactly 16; under a monotone response, an unobserved integer threshold can lie between 9 and 16. Without monotonicity, even this interpolation is not justified.

2. **Practical plateau:** Whether the budget-response curve has flattened enough that further budget increases yield negligible improvement. This requires a separate plateau criterion (e.g., $|A(2m) - A(m)| < \delta$) and is not automatically established by m_c.

3. **Asymptotic threshold:** The threshold relative to an asymptotic performance level $A_\infty = \lim_{m \to \infty} A(m)$, if this limit can be identified. Our data do not identify a stable asymptote for all families; we therefore report grid thresholds, not asymptotic thresholds.

Values reaching the upper boundary of the sweep ($m_c = 64$) indicate that the threshold was not reached at a lower budget on the grid. We label these as *boundary cases* rather than "right-censored," because the threshold relative to $A_{\max}^{\text{obs}}$ is always defined on the grid; what is unknown is whether the curve would continue to rise beyond $m = 64$.

An audit across 4 families × 5 topologies × 3 traffic levels × 3 network sizes × 7 budgets × 5 seeds (6,300 nominal runs, of which 5,880 completed; 420 runs missing due to disconnected RGG graphs; 840 reported m_c estimates at $\alpha = 0.95$; frozen file: `v05_fixes/results/p1_mc_values.csv`) shows:

- $m_c = 1$: 0 regimes (0%)
- $m_c \leq 4$ (interior): 187 regimes (22.3%)
- $4 < m_c < 64$ (interior): 355 regimes (42.3%)
- Boundary case ($m_c = 64$): 298 regimes (35.5%)

Table 4 shows the per-family m_c distribution.

Table 4: Per-family m_c at $\alpha = 0.95$ (per-node budget). "Boundary" indicates $m_c = 64$.

Family	Median m_c	Range	Boundary	n
Trapped Ion	4	[2, 16]	0/210	210
Neutral Atom	16	[4, 64]	52/210	210
Diamond (Electron)	32	[8, 64]	102/210	210
AFC	64	[2, 64]	144/210	210

The threshold location varies strongly by memory profile: the Trapped Ion profile reaches the 95%-of-observed-maximum threshold at a median $m_c = 4$, while AFC reaches the boundary ($m_c = 64$) in 69% of its regimes. Boundary observations are valid grid thresholds, not censored values; we report medians to describe their discrete distribution. The data support a regime-dependent threshold, not a universal constant. These m_c estimates are model-specific quantitative results of the WestQuant simulation; a coarser cross-implementation sweep reproduced the qualitative regime-dependence of the memory threshold but showed sensitivity of threshold location to implementation details. Direct quantitative cross-validation of the full $m = \{1, 2, 4, 8, 16, 32, 64\}$ threshold analysis remains future work.

α **stability:** The median m_c increases from 16 ($\alpha = 0.90$) to 32 ($\alpha = 0.95$ and 0.99), indicating that the threshold is moderately sensitive to the saturation criterion.

7 Failure of Universal Scalar Scaling

We test four dimensionless scaling hypotheses. The first three are motivated by dimensional analysis of the load–capacity ratio and its topology corrections; the fourth incorporates memory physics:

$$\Pi = \frac{\lambda\, t_{\text{attempt}}}{m\, p_{\text{gen}}} \tag{9}$$

$$\Pi^* = \Pi \cdot \sqrt{\bar{h}} \tag{10}$$

$$\Pi_B = \max_v \rho_v, \quad \rho_v = \frac{\Lambda_v}{m_v \cdot \mu}, \quad \mu = \frac{p_{\text{gen}}}{t_{\text{attempt}}} \tag{11}$$

$$\Pi_{\text{phys}} = \frac{\Pi}{\eta_r \cdot \exp(-t_{\text{storage}}/T_2)} \tag{12}$$

where $\bar{h}$ is the mean shortest-path length, Λ_v is the traffic-weighted betweenness of node v, m_v is the memory at node v, and t_{storage} is the mean storage time. Π_B is the maximum node

utilization (bottleneck-flow parameter). Π_{phys} incorporates retrieval efficiency η_r and coherence decay.

Quantitative test: We compute the coefficient of determination R^2 for A_{req} as a function of each scalar (log-transformed). All four candidates are evaluated in the pooled data; Π, Π_B, and Π_{phys} are also evaluated separately within each memory profile (Table 5):

Table 5: R^2 of A_{req} versus the three candidates evaluated per profile. Π^* was evaluated in the pooled global collapse test ($R^2 = 0.042$) but was not separately refit per profile in the frozen closure analysis.

Profile	$R^2(\Pi)$	$R^2(\Pi_B)$	$R^2(\Pi_{\text{phys}})$
Trapped Ion	0.153	0.255	0.245
Neutral Atom	0.081	0.176	0.166
Diamond (Electron)	0.040	0.123	0.112
Diamond (Nuclear)	0.024	0.103	0.085
AFC	0.017	0.084	0.067
Superconducting	0.007	0.042	0.000
Pooled R^2	0.025	0.052	0.177

Among the three candidates evaluated per profile, no scalar achieves $R^2 > 0.26$; all four candidates have pooled $R^2 < 0.18$. For Π alone, 85–99% of within-profile response variance remains unexplained, and Figure 3 shows that the residual structure is visibly organized into topology-specific bands. Table 6 quantifies the unexplained fraction per profile.

Table 6: Fraction of within-profile response variance unexplained by the log-Π regression, computed as $1 - R^2(\Pi)$. This quantity is residual variance, not a direct variance-component estimate for topology.

Profile	Unexplained fraction $1 - R^2(\Pi)$
Trapped Ion	0.847
Neutral Atom	0.919
Diamond (Electron)	0.960
Diamond (Nuclear)	0.976
AFC	0.983
Superconducting	0.994

Figure 3 shows the scalar collapse test, and Table 5 compares the R^2 values across families.

Claim: In the WestQuant dataset, none of the four tested scalar reductions achieves a topology- and hardware-independent collapse. We do not claim that no universal scaling law can ever exist—only that these specific candidates are insufficient. The physics-aware candidate Π_{phys} achieves the highest overall R^2 (0.177) but still leaves more than 82% of variance unexplained. The scalar-collapse result has not yet been independently cross-implemented.

Nonlinear collapse test: Because the linear R^2 test may miss a nonlinear monotonic relationship, we additionally compute (i) Spearman rank correlation ρ_s between each scalar and A_{req}, and (ii) R^2 of a quadratic log-transformed fit $A_{\text{req}} = a(\log \Pi)^2 + b \log \Pi + c$. Table 7 reports the pooled results. For Π, Π^*, and Π_B, the quadratic R^2 remains below 0.06 and $|\rho_s| < 0.22$, confirming that both linear and nonlinear monotonic fits fail. For Π_{phys}, the quadratic fit improves to $R^2 = 0.379$ (pooled) with $\rho_s = -0.721$, but per-family quadratic R^2 remains below 0.28 (Trapped Ion: 0.274; AFC: 0.069), indicating that the pooled improvement is driven by inter-family separation rather than within-family collapse. We therefore conclude that the failure of universal scalar scaling is robust to nonlinear monotonic transformations for Π, Π^*, and Π_B, and that Π_{phys} shows partial nonlinear improvement but does not achieve a topology-independent within-profile collapse.

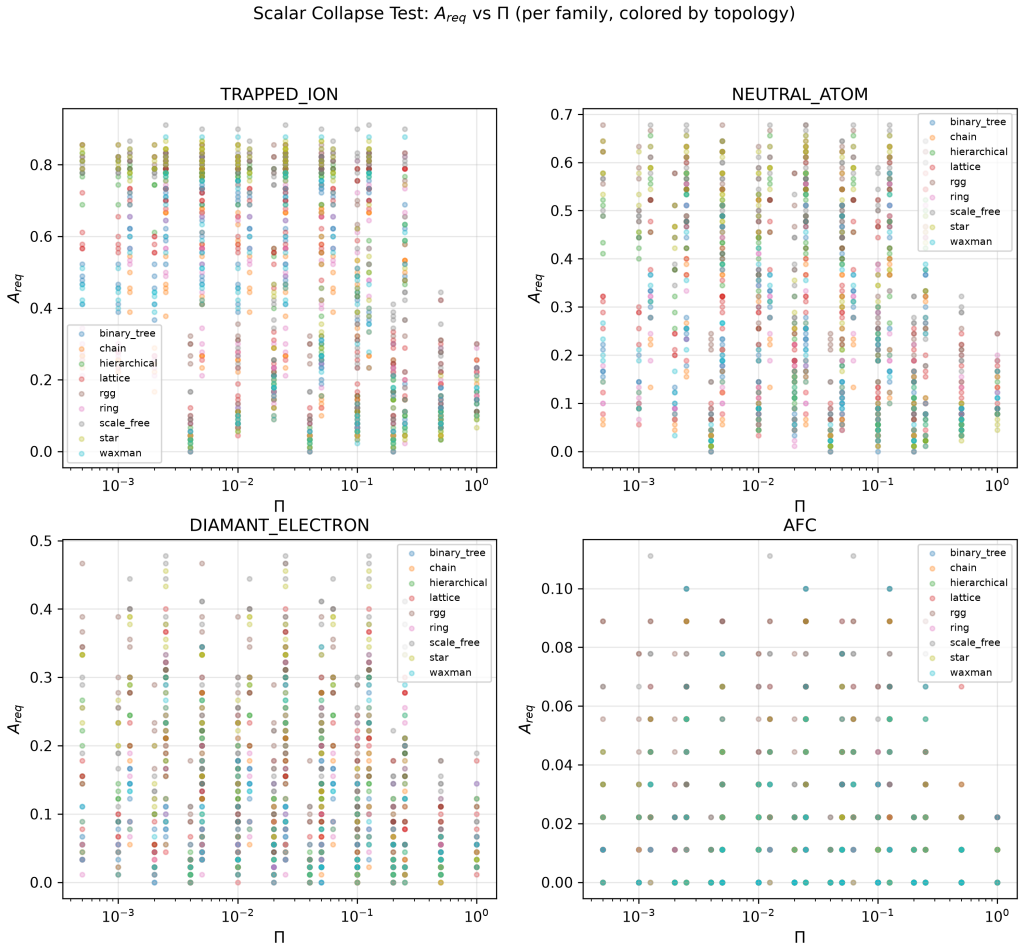


Figure 3: Scalar collapse test: A_{req} vs Π for four memory families, colored by topology. Topology-specific bands persist within each family, confirming that Π does not capture topology-dependent variation even when hardware is held fixed.

Table 7: Nonlinear scalar-collapse test (pooled). R^2_{lin}: linear log-transformed. R^2_{quad}: quadratic log-transformed. ρ_s: Spearman rank correlation.

Scalar	R^2_{lin}	R^2_{quad}	ρ_s
Π	0.025	0.029	-0.118
Π^*	0.042	0.045	-0.169
Π_B	0.052	0.058	-0.214
Π_{phys}	0.177	0.379	-0.721

8 Retrieval-Window Descriptor and Limits of Analytical Sizing

The scalar-collapse analysis models A_{req}, whereas memory sizing targets m_c; fit statistics for these different outcomes are not directly comparable. A scalar containing the candidate budget m is not inherently circular when modeling $A(m)$. Circularity would arise if the unknown target m_c itself were supplied as a predictor of m_c.

8.1 An independent-channel reference calculation

For a hard window of length τ_r, let $K = \lfloor \tau_r / t_{\text{attempt}} \rfloor$ independent Bernoulli attempts occur on one link. Then

$$p_w = 1 - (1 - p_{\text{gen}})^K, \qquad q_{\min} = \left\lceil \frac{\ln(1-\alpha)}{\ln(1-p_w)} \right\rceil. \tag{13}$$

Here $q_{\min}$ counts independent channels with success probability p_w, and targets the absolute probability $1 - (1 - p_w)^q \geq \alpha$. It is not the per-node budget threshold relative to the observed maximum. Moreover, τ_r in the simulator is an exponential decay constant, not a deterministic expiry time. The hard-window calculation is therefore an illustrative reference, not an exact model of retrieval survival.

Table 8: Retrieval-window descriptor at $p_{\text{gen}} = 0.005$ and $t_{\text{attempt}} = 10^{-3}$ s. The illustrative independent-channel target uses absolute $\alpha = 0.95$; the final column uses the distinct empirical relative-grid threshold.

Profile	χ_r	p_w	$q_{\min}$	Median m_c
Trapped ion	50	≈ 1	1	4
Neutral atom	2.5	0.9184	2	16
Diamond electron	0.5	0.3942	6	32
AFC	0.05	0.0489	60	64

For h independent links required to succeed within a common hard window, a channel succeeds with probability p_w^h. Replacing p_w by p_w^h in the expression for $q_{\min}$ requires independent complete path channels, which are not supplied automatically by m slots per node. For AFC and $h = 2$, this idealization gives about 1,252 channels, illustrating why it must not be presented as a validated predictor of grid thresholds bounded by 64. No predictive accuracy or invariance rate is asserted for the corrected χ_r in this revision.

8.2 Physical interpretation

Small χ_r indicates few expected link successes during a retrieval decay time. Large χ_r reduces this particular timescale constraint but does not establish queue-limited operation: offered load, service time, memory occupancy, deadlines, routing, and swap success must also be considered. Coherence time can influence fidelity and A_{qual} even when it has no direct role in a completion-only acceptance rule. Claims of zero causal influence require an independent-parameter experiment with the actual acceptance semantics specified.

9 Heterogeneous Memory Placement

9.1 Heterogeneity Reconciliation

We resolve an apparent contradiction in earlier exploratory results with a controlled factorial experiment over 6 topologies, 2 network sizes, 3 traffic levels, 3 coupling rules, and 5 seeds. Four mixed-placement strategies are evaluated at both high-quality fractions, while the two homogeneous controls (all-high and all-low) are evaluated once per graph/traffic/coupling condition.

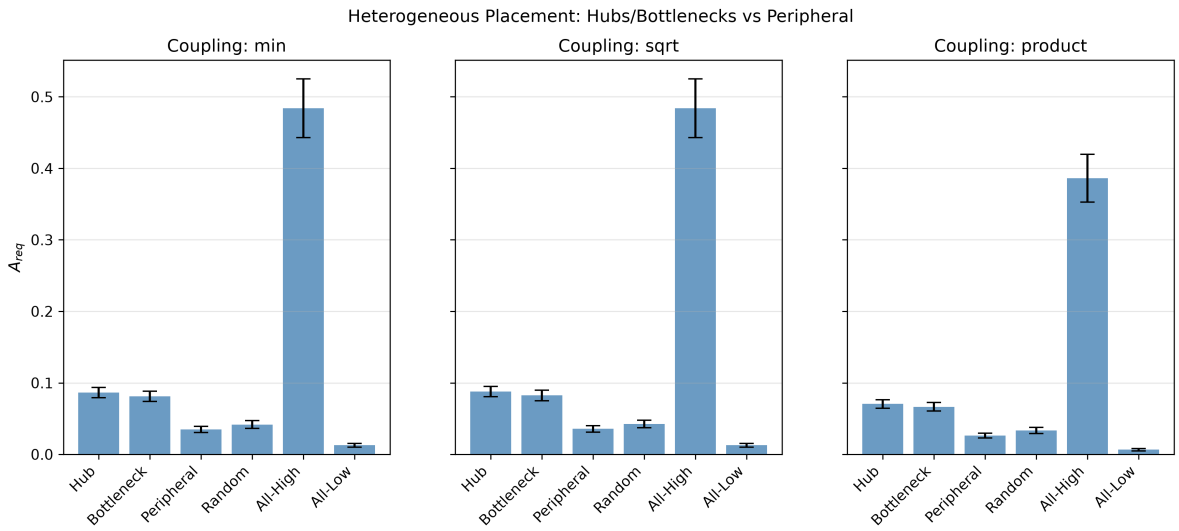


Figure 4: Heterogeneous placement results across three coupling rules. Aggregated across the tested configurations, hub and bottleneck placement outperform peripheral placement, although the per-topology ranking is not universal.

This gives 5,400 nominal runs. Ninety runs (1.7%) are skipped due to disconnected RGG graphs at $N = 20$, leaving 5,310 actual runs.

Definitions: *Hub* nodes are the top-k nodes by degree centrality. *Bottleneck* nodes are the top-k nodes by betweenness centrality. *Peripheral* nodes are the bottom-k nodes by degree centrality. In a star topology, the hub and bottleneck nodes coincide.

Result: Hubs (mean $A = 0.0815$) and bottlenecks (mean $A = 0.0766$) outperform peripheral placement (mean $A = 0.0323$). The strategy ranking is robust across coupling rules (Kendall $\tau = 1.0$ for all pairs of {min, sqrt, product}).

Paired comparison (hub vs bottleneck): Across 1,062 matched pairs, the median paired difference is 0 (558 ties, 322 hub wins, 182 bottleneck wins). A Wilcoxon signed-rank test is significant ($p < 0.001$) due to the asymmetry in non-tie outcomes, but the effect size is small (mean difference = 0.005, Cohen's $d = 0.19$). The Wilcoxon test uses the standard *wilcox* zero method, which discards the 558 zero-difference pairs before ranking, leaving 504 non-zero pairs; the p-value is computed via the normal approximation. A reported two-one-sided-tests (TOST) equivalence procedure with margin $\delta = 0.02$ supports equivalence at the chosen margin ($p_{\text{TOST}} < 0.001$), indicating that the mean paired difference is within ± 0.02 on the A_{req} scale. We conclude that hub and bottleneck placement are *statistically distinguishable but practically equivalent* in the tested configurations, supported by both the small effect size and the formal TOST result.

Per-topology ranking: The ranking is not perfectly consistent across topologies. In chain, peripheral outperforms bottleneck; in lattice, bottleneck outperforms hub. The overall ranking (hub > bottleneck > random > peripheral) holds in 4 of 6 topologies. Figure 4 shows the heterogeneous placement results across coupling rules.

9.2 Interface-Coupling Model Sensitivity

The three coupling rules (min, $\sqrt{\eta_i \eta_j}$, $\eta_i \eta_j$) are alternative model assumptions used to test sensitivity to the physical interface between heterogeneous memories at the two endpoints of a link. They produce different absolute performance levels but preserve the ranking of placement strategies. The min and $\sqrt{\cdot}$ rules produce similar absolute levels (within 2%), while the product rule lowers absolute A_{req} by 18–25% depending on strategy. These coupling rules are explicit modeling assumptions and are not native concepts in the external frameworks used in the

cross-implementation study; they have not been independently validated. This confirms that heterogeneous-memory conclusions require explicit interface modeling.

9.3 Gap Between Mixed and All-High Placement

A notable finding is the large gap between any mixed placement and the all-high baseline: mixed strategies recover only 8–18% of the all-high performance. This gap describes the tested parameter bundles and fractions; it is not a universal cost of heterogeneous hardware.

10 Global Sensitivity Analysis

10.1 Permutation Importance Cross-Check

As an independent sensitivity cross-check, we compute permutation importance using a Random Forest surrogate trained on the 67,584 A08 samples. Permutation importance measures which variables the surrogate is most dependent on for its predictions; it does not directly measure causal physical importance. The surrogate's held-out R^2 and MAE are reported alongside the importance values to contextualize their reliability (Table 9).

Table 9: Random Forest surrogate performance on held-out data (20% test split, 200 trees). Per-topology R^2 is high; the lower overall R^2 reflects topology as a major driver not included as a feature.

Topology	n	R^2	MAE
All (pooled)	67,584	0.423	0.142
Chain	22,528	0.944	0.015
Star	22,528	0.972	0.019
RGG	22,528	0.974	0.023

The per-topology held-out R^2 values (0.94–0.97) indicate good prediction within the sampled design. Permutation importance describes predictive association, with dependence on the feature distribution and split; high R^2 alone does not validate every importance estimate. The lower pooled $R^2 = 0.42$ is consistent with omitted topology information, but this explanation requires a matched surrogate comparison to isolate. Table 10 lists the parameter ranges, and Figure 5 shows the permutation importance by topology.

Table 10: Parameter ranges for the sensitivity analysis. Log-scaled parameters are sampled uniformly in log-space.

Parameter	Min	Max	Scale
η_r (retrieval eff.)	0.30	0.99	linear
F_{ro} (readout fid.)	0.90	0.999	linear
T_2	0.001 s	10 s	log
p_{gen}	0.001	0.05	log
t_{attempt}	10^{-5} s	10^{-2} s	log
m (memory budget)	1	8	discrete
λ (arrival rate)	1.0	100.0	log
noise scale	0.1	10.0	linear
N (network size)	10	50	discrete
F_{initial}	0.85	0.99	linear

Result: Retrieval efficiency is consistently ranked as the most informative predictor by the Random Forest surrogate across all three tested topologies (permutation importance 0.78–1.15),

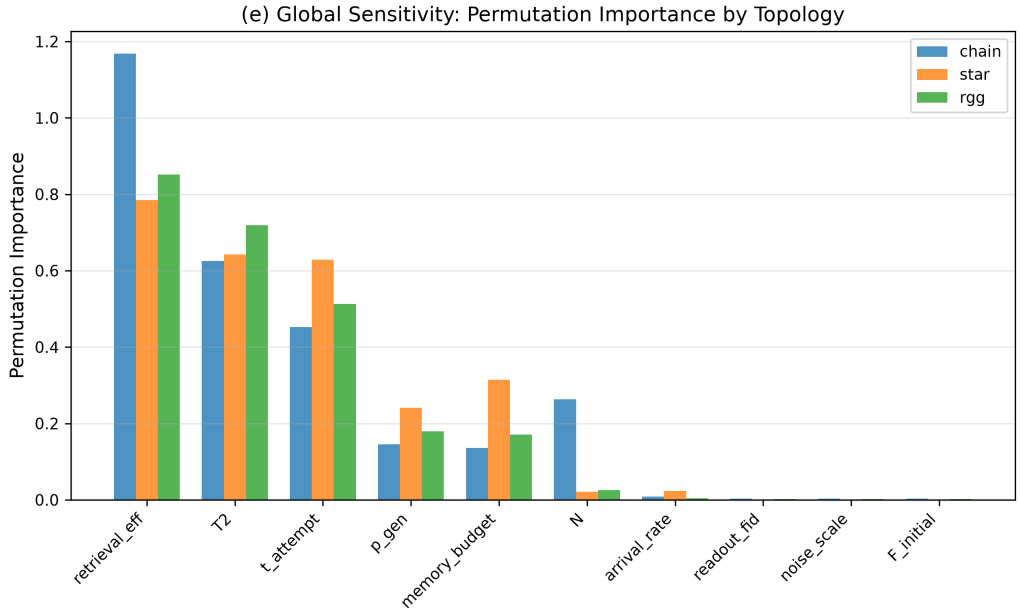


Figure 5: Permutation importance by topology. Retrieval efficiency is consistently the most important parameter across all three topologies, followed by T_2 and t_{attempt}.

followed by T_2 (0.61–0.67) and t_{attempt} (0.40–0.59). The ordering is consistent across the three tested topologies.

Caveat: Retrieval efficiency η_r enters the success probability multiplicatively, so its high importance is partly structural rather than purely physical. The importance of η_r should be interpreted as "most informative for the surrogate's predictions," not as "causally dominant in the physical system." The broad range of η_r (0.30–0.99) also contributes to its high ranking; a narrower range would likely reduce its importance. The feature-importance ordering has not yet been evaluated across independent simulation models.

Interpretation of the T_2 importance. The surrogate uses the 67,584-sample A08 design, not the six-profile atlas. Rank correlation between T_2 and τ_r in the atlas cannot by itself explain this separate sensitivity result. The parameter-to-decay mapping and response variable in the frozen A08 implementation must be checked before assigning a causal interpretation. Permutation importance remains a predictive association within that design.

11 Design Implications

The observed memory threshold should be reported together with its budget grid, metric, traffic, topology, and memory profile. A boundary value of 64 is a finite-grid result, not a measured asymptote. At fixed N and m, total storage is matched across topologies; path length, connectivity selection, and routing still influence the comparison.

The parameter χ_r is useful for identifying a retrieval timescale relative to generation, but it does not replace queueing or multi-hop analysis. Memory-profile rankings are comparisons of adopted parameter bundles rather than certification of specific devices. In heterogeneous deployments, hub and betweenness-based placement are promising choices under the tested assumptions, with documented topology-specific exceptions. The TOST margin of 0.02 is substantial relative to the aggregate completion fractions of approximately 0.08; equivalence at that margin should not be interpreted as interchangeability for every application.

12 Limitations

1. **Cross-implementation rather than full native-protocol validation.** In addition to analytical component checks, selected conclusions were tested through harmonized implementations using SeQUeNCe and SimQN adapters. The two external adapters share a common harmonized model and therefore do not constitute two independent physical-model replications. Native SeQUeNCe/SimQN protocols, NetSquid, and QuISP were not included.

2. **Absolute-value model dependence.** Cross-implementation comparisons indicate that absolute performance values are more implementation-sensitive than the tested topology and memory-profile orderings.

3. **Critical memory capacity.** Numerical critical-memory thresholds have not yet been cross-validated using the full $m = \{1, 2, 4, 8, 16, 32, 64\}$ sweep.

4. **Heterogeneous placement.** The heterogeneous placement result has not yet been cross-implemented.

5. **Interface-coupling rules.** The interface-coupling rules are explicit modeling assumptions and have not been independently validated.

6. **Metric semantics.** Completion and latency metrics are sensitive to queueing, censoring, and timing definitions; future native-protocol comparisons must harmonize these definitions before quantitative comparison.

7. **Scalar collapse.** The scalar-collapse result has not yet been independently cross-implemented.

8. **Sobol bootstrap bug.** The Saltelli bootstrap confidence intervals were found to be unreliable due to a structural resampling bug (resampling individual rows instead of Saltelli blocks). Sobol indices are therefore not reported as quantitative results; fixing the bootstrap and rerunning is future work. We rely on permutation importance for qualitative ranking.

9. **Representative memory parameters.** Memory families use representative parameter bundles, not complete hardware-stack comparisons. Real devices have additional noise sources.

10. **Uniform traffic in main atlas.** Most atlas experiments use uniform traffic. Traffic-matrix support exists but is not exercised in the main atlas.

11. **No purification or multiplexing.** The simulator does not model entanglement purification or spatial multiplexing.

12. **Limited network sizes.** Experiments cover $N \in [10, 50]$. Scaling to $N \geq 100$ is future work.

13. **Routing policy.** We use shortest-hop routing. Other policies may change conclusions.

14. **No dynamic traffic.** Traffic is Poisson with fixed rate. Time-varying traffic belongs to future work.

15. **Atlas and confirmatory grid completeness.** The atlas design contains 9,720 nominal cells; 9,360 completed runs are retained (360 missing, 3.7%). The confirmatory grid contains 9,000 cells; 8,730 completed simulations are retained (270 missing, 3.0%). Missing cells arise from disconnected random-geometric graphs and simulation failures and are not imputed.

16. **Topology fairness.** Equal per-node budgets give equal total storage at fixed N. Pooling different network sizes requires stratification; path length, edge density, and connectivity selection remain relevant at fixed size.

17. **Topology-dependent exclusion of no-path requests.** The metric A_{req} excludes no-path requests from the denominator. Because the fraction of no-path requests depends on topology (e.g., disconnected RGGs produce more no-path requests than fully connected chains), the effective denominator varies across topologies, potentially biasing cross-topology comparisons.

18. **Seed and request quantization.** The atlas uses 5 seeds and 100 requests per simulation. With at most 100 effective requests, the resolution is $1/N_{\text{effective}}$ and can be coarser than 0.01 after exclusions. The confirmatory dataset uses 20 seeds, which reduces but does not eliminate this quantization effect.

19. m_c **relative to observed maximum.** The threshold m_c is defined relative to $A_{\text{max}}^{\text{obs}}$, the maximum on the observed budget grid, not an asymptotic limit. If the budget-response curve has not plateaued, m_c underestimates the budget needed to reach a fixed fraction of the true asymptotic performance. This is a fundamental limitation of grid-based threshold estimation.

20. **RGG disconnection selection bias.** Disconnected random-geometric graphs are excluded from the analysis (contributing to the 360 missing atlas cells and 270 missing confirmatory cells). This creates a selection bias: the surviving RGG instances are those with sufficient edge density to remain connected, which may overstate RGG performance relative to a random draw from the RGG ensemble.

21. **Readout-model versioning.** Reported results use the frozen multiplicative readout rule. Later depolarizing readout changes fidelity by $(1 - F_{ro})/4$ and may change fidelity-qualified results; these versions must not be mixed.

22. **Run length at high traffic.** With $n_{\text{requests}} = 100$ and $\lambda = 50$, the observation window is approximately 2 s, which may not reach full stationarity for queueing dynamics at high load. Results at $\lambda = 50$ should be interpreted with this caveat.

13 Conclusion

Memory capacity requirements in the tested quantum-network model depend on memory profile, topology, and traffic. The finite-grid threshold reaches the upper tested budget in 35.5% of the reported regimes, and it must be distinguished from an asymptotic saturation capacity. The tested scalar reductions leave substantial topology-dependent variation. Correctly using retrieval decay time in χ_r provides a physical timescale descriptor but does not establish a universal sizing law. Heterogeneous placement favors hubs and betweenness-based nodes in aggregate, while topology-specific exceptions and coupling-dependent absolute performance limit generalization. These findings support explicit, model-specific capacity planning and targeted validation of the decisions to be deployed.

Data Availability and Reproducibility

The manuscript and a repository scaffold are prepared for a dedicated public reproducibility release. The raw simulation datasets, frozen simulator, dependency lockfile, complete seed/configuration ledger, and figure-generation scripts have not yet been deposited in that release. The development repository is private. This revision therefore does not claim that the empirical results can already

be reproduced from a public archive. A public release URL, commit identifier, checksums, and archival DOI should identify the completed deposit before submission.

Relationship to Companion Work

A companion methods manuscript, "Assessing Decision Reliability in Discrete-Event Simulation: A Quantum-Network Case Study," studies stochastic decision stability, metric semantics, and model sensitivity. The present article focuses on memory-profile, capacity, topology, and placement results. Shared simulator infrastructure and overlapping validation results should be disclosed to both editors.

Competing Interests

The author is the founder of Vesterlund Ventures and WestQuant Open. The simulator used in this study (WestQuant) was developed by the author. This creates a potential conflict of interest insofar as the simulator's design choices may influence the results. The cross-implementation robustness check (Sec. 4.3) partially mitigates this concern by testing whether selected conclusions survive a harmonized model pathway with shared code, but does not eliminate it. The author has no external funding to declare.

Author Contributions

D.V. conceived the study, designed and implemented the simulator, ran all experiments, performed all analyses, and wrote the manuscript.

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